

The processed dataset contains 29 columns that appear in all 20 years. This count includes the diabetes target, year, and some related versions of the same information, so it does not mean that all 29 columns will be used as predictors. Most columns have similar categories across years, but changes in the original CDC questions and preprocessing still need to be considered.

For example, high blood pressure and high cholesterol are only available in odd-numbered years, as documented by the dataset author. Income also changes in 2021 from 8 income groups to 11 groups. The source code uses different versions for these variables to represent this change.

I also checked the diabetes target. The raw CDC diabetes variable changes across the years (DIABETE2, DIABETE3, and DIABETE4), but the Kaggle preprocessing uses the same three-class mapping: Diabetic, Non-diabetic, and Pre-diabetic. For example, the 2005 processing code maps the original responses into these three classes and removes records without a usable diabetes response.

Two issues still need more investigation. `race_00` normally has 8 categories, but the "Other" category is absent in the 2022 processed data. Also, `employment_status_00` is completely missing in the 2024 processed data, although it is expected to have 8 categories. The feature documentation describes both variables with 8 categories.

My next step is to compare the important variable changes with the official CDC BRFSS documentation. I will focus on changes in the diabetes variable, race, employment, income, and other variables whose original CDC variable names changed over time. This will help determine whether the yearly data can be compared consistently before building the temporal models.

2011 BRFSS Methodology Change: CDC identifies 2011 as a new baseline because of changes in survey weighting and the addition of cell-phone respondents. Therefore, changes around 2011 will be interpreted carefully. The 2011 transition will also be examined as a possible source of distribution shift in the temporal robustness analysis.

Diabetes: The raw CDC variable changes from DIABETE2 to DIABETE3 and later DIABETE4. CDC documentation shows that the main diabetes question and response coding remain comparable across these changes. The Kaggle preprocessing also applies a consistent three-class mapping across years. Therefore, the processed diabetes target can be used for temporal comparison, while documenting that pregnancy-only diabetes is grouped with the Non-diabetic class.

Race: Race is generally represented using eight categories in the processed dataset. In 2022, the source variable changes from _RACE to _RACE1, and the processed data do not contain the "Other" category. CDC documentation confirms that the 2022 calculated race variable was constructed differently. I will treat 2022 as an exception and consider its effect when interpreting temporal changes.

Employment: Employment status is available in the earlier processed files, but all 454,774 values are missing in the processed 2024 file. The CDC 2024 variable layout still lists EMPLOY1, so I cannot yet determine why the Kaggle-processed column is completely missing. I will check the preprocessing code and exclude employment from the main common feature set if the 2024 values cannot be recovered.

Income: CDC changed the income variable from INCOME2 to INCOME3 in 2021. The newer variable expands the highest income category into four groups. For temporal consistency, I plan to combine these newer high-income groups into the older \$75,000 or more category. This will create a consistent 8-level income variable across 2005–2024.

High blood pressure: This variable is available only in odd-numbered survey years. CDC documentation confirms that the underlying blood-pressure variable changes version over time, but the question and main response categories remain comparable. Because the feature is unavailable in even years, it will not be used in the common 2005–2024 feature set, but it may be considered in a separate odd-year analysis.

High cholesterol: The underlying variable changes from TOLDHI2 to TOLDHI3, while the main question and response categories remain comparable. The variable is available only in selected survey years and also contains structural missingness because respondents

who have not had their cholesterol checked are not asked whether they were told it was high. Therefore, this feature will not be included in the common 2005–2024 feature set but may be used in an odd-year sensitivity analysis.

Sex: The underlying BRFSS variable changes from SEX to SEX1 and later to SEXVAR. CDC documentation shows that these variables continue to represent respondent sex using male/female categories, although the construction of the variable changes over time and SEXVAR incorporates survey-mode-specific source variables. Therefore, the processed sex feature can be retained for temporal analysis, with the variable-source change documented.

Alcohol consumption: The calculated alcohol variable changes from DRNKANY4 to DRNKANY5 and later to DRNKANY6. CDC documentation shows that these version changes follow changes in the underlying alcohol-frequency variables, while the calculated feature continues to represent whether the respondent consumed alcohol during the past 30 days. Therefore, the harmonized binary alcohol-consumption feature can be retained for temporal analysis.

Average drinks per day: The underlying variable changes from AVEDRNK2 to AVEDRNK3 and later to AVEDRNK4, while CDC documentation indicates that the main concept remains average alcohol consumption on drinking days during the past 30 days. The high missing rate is largely structural because respondents who report no recent alcohol consumption are not asked this question. Therefore, missing values should not be treated using simple generic imputation.

Heart attack and coronary heart disease: The underlying BRFSS variables change from CVDINFR3/CVDCRHD3 to CVDINFR4/CVDCRHD4 beginning in the later survey years. CDC documentation shows that the variables continue to represent self-reported history of heart attack and coronary heart disease, and the version 4 variables remain in use through 2024. Therefore, both harmonized features can be retained for temporal analysis.

Stroke: The BRFSS source variable CVDSTRK3 remains consistent across the study period, and no variable-version change was identified in the preprocessing audit. The feature

represents self-reported history of stroke and can be retained in the common temporal feature set.

Cost barrier: The BRFSS source variable changes from MEDCOST to MEDCOST1 beginning in 2021. CDC documentation shows that the feature continues to measure whether the respondent needed to see a doctor during the past 12 months but could not because of cost. Therefore, the harmonized cost-barrier feature can be retained for temporal analysis.

Personal health care provider: The source variable changes from PERSDOC2 to PERSDOC3 in 2021. The newer question explicitly includes a group of doctors. In the processed data, the percentage answering “More than one” increases from 6.81% in 2020 to 28.35% in 2021. I will document this change and compare model results with and without this feature before deciding whether to keep it in the main analysis.

Last routine checkup: The source variable changes from CHECKUP to CHECKUP1, but CDC documentation shows that the question concept and response categories remain comparable. The feature continues to measure the length of time since the respondent's last routine medical checkup. Therefore, the harmonized feature can be retained for temporal analysis.

Weight, height, and BMI: The underlying BRFSS weight (WEIGHT2) and height (HEIGHT3) variables remain consistent across the study period. CDC also uses these variables to calculate BMI. However, the Kaggle dataset calculates BMI during preprocessing rather than directly using the CDC calculated BMI variable. CDC applies additional validity and pregnancy-related exclusions when calculating BMI. Therefore, the Kaggle BMI calculation will be verified against CDC rules before modeling.

Age: The calculated BRFSS variable _AGEG5YR is used throughout the study period. CDC documentation consistently defines it as age 18–24 followed by five-year age groups through age 80 or older. Therefore, the age-group feature can be retained for temporal analysis without additional harmonization.

Education: The BRFSS source variable EDUCA remains consistent throughout the study period. CDC documentation also uses consistent education categories and category groupings across the years reviewed. Therefore, the education feature can be retained for temporal analysis without additional harmonization.